GraphOne AI Intelligence Pipeline Architecture

Page 1 — System Overview

Data Sources: arXiv, Greenhouse API, RSS Feeds, AI Company Websites.

Scraper Layer: Python requests, aiohttp, feedparser, BeautifulSoup.

LLM Layer: Gemini Flash → Groq Llama 3 → DeepSeek fallback chain.

Entity Resolution Layer: Canonical mapping and deduplication.

Output Layer: CSV files, PostgreSQL, Neo4j, Google Sheets.

Page 2 — Scalability Strategy

The crawler is horizontally scalable using asynchronous workers.

Massive crawling uses pagination and concurrent workers.

429 Handling: Exponential Backoff (2s, 4s, 8s, 16s).

413 Handling: HTML is chunked into semantic blocks before LLM extraction.

Page 3 — Freshness & Storage

Freshness Tracking uses source URL + publication timestamp to avoid duplicates.

Entity Resolution converts inconsistent startup names into canonical entities.

PostgreSQL stores structured entities, Neo4j stores relationships, and vector storage enables semantic retrieval.